

Ushtrime ND2b

1.1.4 Trupi fillon të lëvizë me shpejtësi fillestare $v_0 = 10 \text{ m/s}$, nëpër një rrugë

drejtëvizore me nxitim konstant $a = 2 \text{ m/s}^2$.

Sa shpejtësi ka trupi pas kohës $t = 50 \text{ s}$?

Sa rrugë kalon trupi gjatë asaj kohe?

Zgjidhje:

(të gjejmë formulën)

Të dhënat:

$$v_0 = 10 \frac{\text{m}}{\text{s}}$$

$$a = 2 \frac{\text{m}}{\text{s}^2}$$

$$t = 50 \text{ s}$$

$$v = ?$$

$$s = ?$$

$$a = \text{konst} > 0$$

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

$$a dt = dv / \int$$

$$\int dv = a \int dt$$

$$v = at + c$$

$$t = 0, v = v_0$$

$$v_0 = a \cdot 0 + c$$

$$v_0 = c$$

$$v = v_0 + at$$

$$v = 10 \frac{\text{m}}{\text{s}} + 2 \frac{\text{m}}{\text{s}^2} \cdot 50 \text{ s}$$

$$v = 10 \frac{\text{m}}{\text{s}} + 100 \frac{\text{m}}{\text{s}}$$

$$v = 110 \frac{\text{m}}{\text{s}}$$

$$v = \text{konst}$$

$$v = \frac{ds}{dt} \cdot dt$$

$$ds = v \cdot dt$$

$$ds = (v_0 + at) dt$$

$$ds = v_0 dt + a dt^2 / \int$$

$$\int ds = v_0 \int dt + a \int t^1 dt$$

$$s = v_0 t + \frac{at^2}{2} + c'$$

$$t = 0, s = 0, c' = 0$$

$$s = v_0 t + \frac{at^2}{2}$$

$$s = 10 \frac{\text{m}}{\text{s}} \cdot 50 \text{ s} + \frac{2 \frac{\text{m}}{\text{s}^2} \cdot 2500 \text{ s}^2}{2} = 500 \text{ m} + 2500 \text{ m}$$

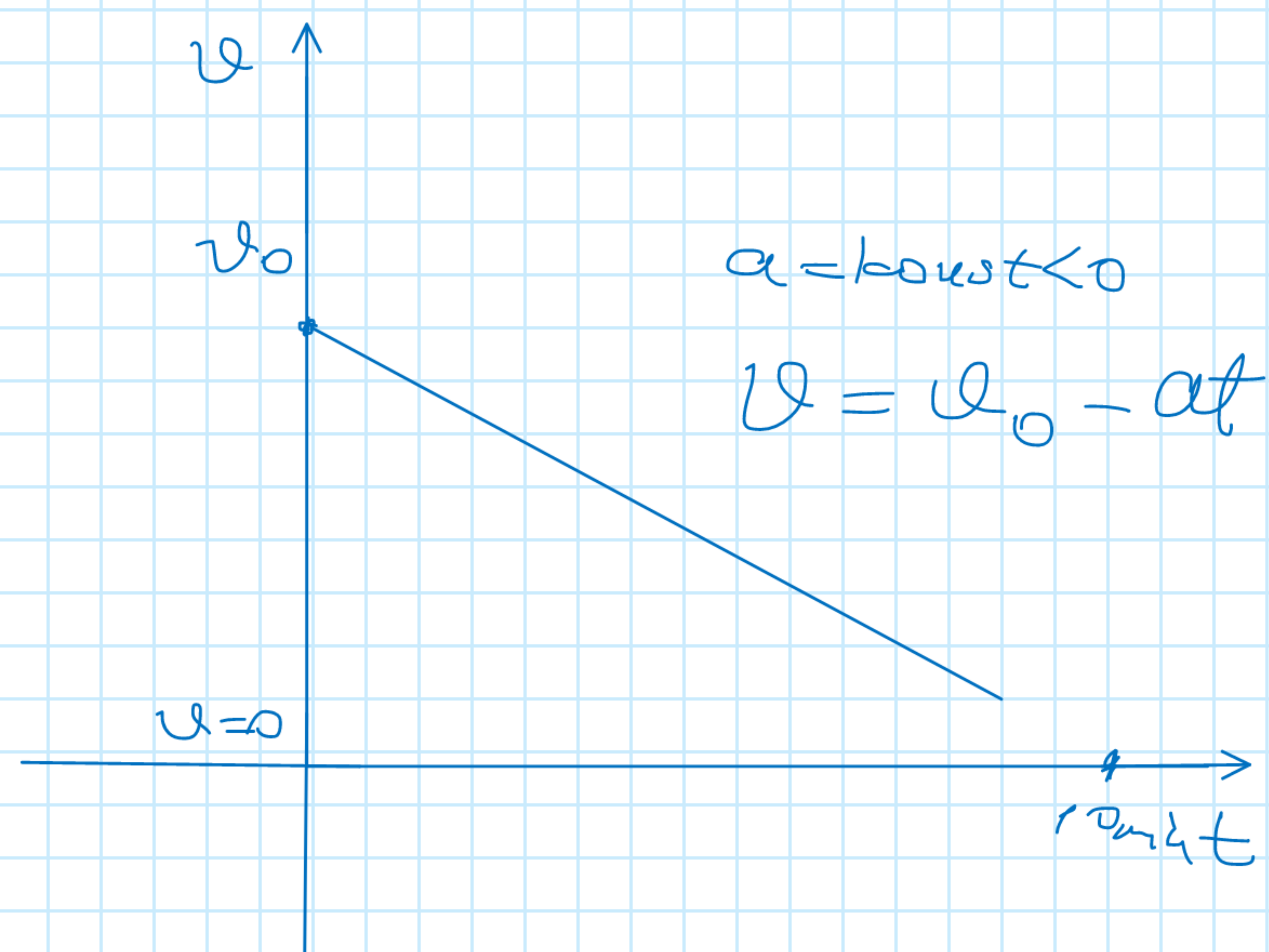
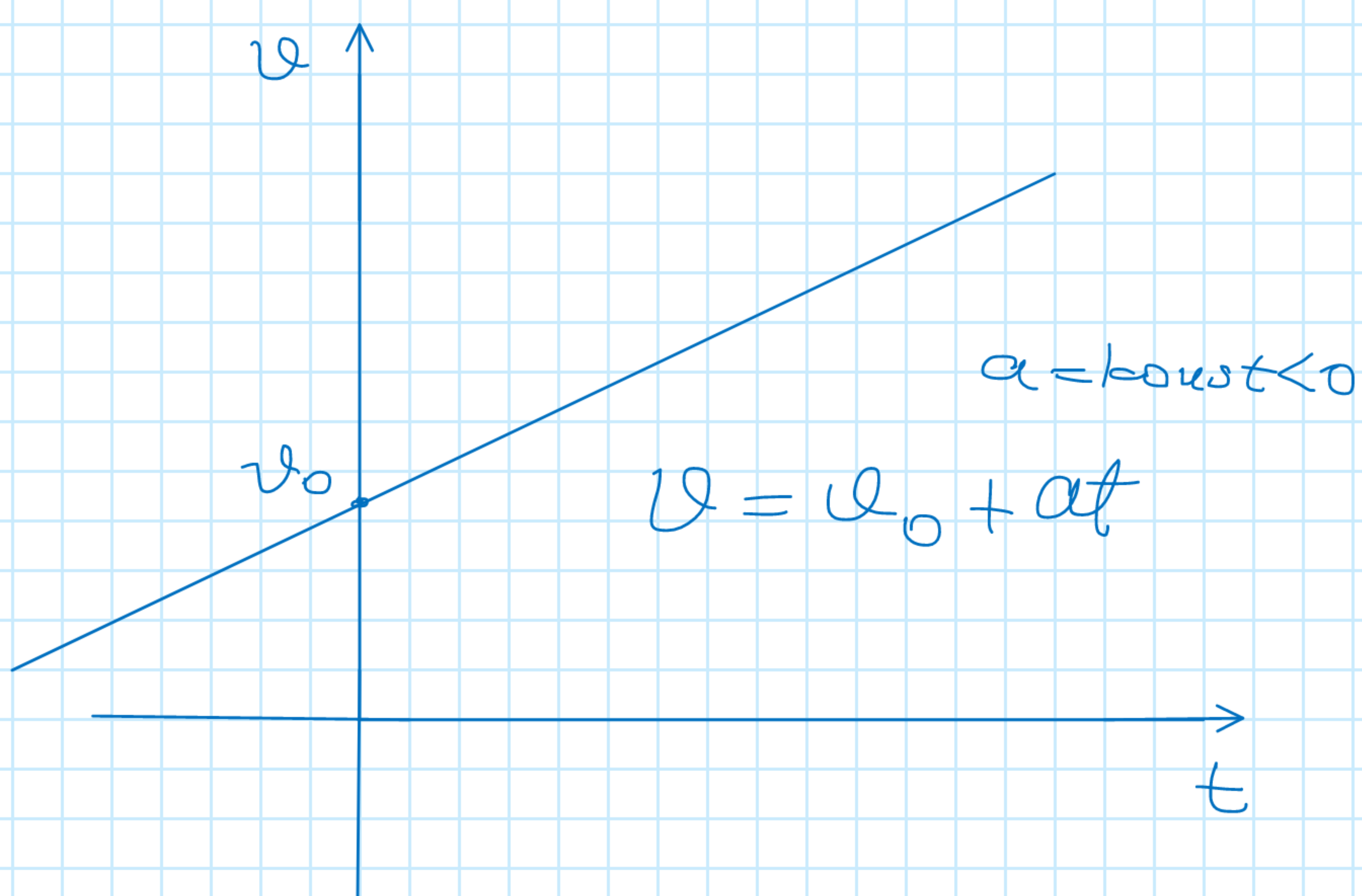
$$s = 3000 \text{ m} = 3 \text{ km} \quad \text{sepse } 1 \text{ km} = 1 \cdot 10^3 \text{ m}$$

$$a_m = \langle a \rangle = \frac{\Delta v}{\Delta t}$$

$$\Delta t \rightarrow 0$$

$$\lim_{\Delta t \rightarrow 0} \langle a \rangle = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

$$a = \frac{dv}{dt}$$



$$\int x^m dx = \frac{x^{m+1}}{m+1} = \frac{x^2}{2}$$

$$m = 1$$

$$\int t^m dt = \frac{t^{m+1}}{m+1}$$

$$m = 1$$